

Collinearity: Diagnosis and Consequences

Part VI gives up unbiasedness in exchange for smaller variance, and collinearity is the situation that first made that trade attractive. When the columns of the model matrix are nearly linearly dependent, the data answer some questions well and others barely at all, and the badly answered ones are often those the analyst asked: the separate effects of regressors that moved together. This chapter says exactly which linear functions lose precision, builds the standard diagnostics (variance inflation factors and their generalization to groups of columns, condition indices and the Belsley–Kuh–Welsch variance decomposition), shows why predictions that follow the pattern of the data survive intact, and ends with the remedies, the last of which, biased estimation, is where [Chapter 27](#) begins.

What you need. [Chapter 1](#) (the spectral theorem and Rayleigh quotients, Theorem 1.7.2 and Theorem 1.7.7; the singular value decomposition and the condition number, Theorem 1.8.1 and Definition 1.8.1), [Chapter 6](#) (the Frisch–Waugh–Lovell theorem and the variance identity Eq. 6.6.2; leverage, Proposition 6.8.1), [Chapter 8](#) (estimability), [Chapter 10](#) (conditioning and column scaling, Theorem 10.5.1 and Proposition 10.5.2) and [Chapter 12](#) (confidence ellipsoids and prediction at a new point). [Section 19.6](#) previewed the subject (Proposition 19.6.1); this chapter builds on that preview rather than repeating it. Four results are used once each: the moments of least squares (Theorem 5.5.1, [Chapter 5](#)) and the criterion for an orthogonal design (Proposition 7.2.4, [Chapter 7](#)) in [Sections 26.1](#) and [26.2](#); the t test of a coefficient (Theorem 11.6.1, [Chapter 11](#)) in [Section 26.2](#); and the deletion formula (Theorem 20.3.1, [Chapter 20](#)) in [Section 26.3](#).

26.-1.1 Roadmap

- [What collinearity does to variance](#)
- [Variance inflation factors](#)
- [Condition indices and variance decomposition](#)
- [Collinearity and prediction](#)
- [What to do about collinearity](#)
- [Summary and notes](#)